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F78AB
Actuarial and Financial Mathematics

Chapter 4: Nominal interest rates

LEADERS IN IDEAS AND SOLUTIONS

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Introducing Nominal Interest Rates

To date our calculations have been based on ***effective*** interest rates

Where, for example, an effective interest rate of 6% per annum is equivalent to an effective interest rate of 2.9563% per half-year.

And more generally, interest of £*i* is paid at the end of a period where £1 is invested at an effective interest rate of *i%* .

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Nominal Interest Rates – new

Nominal interest of $i^{(m)}$ per annum compounding m times a year means that:

$$\frac{£i^{(m)}}{m}$$

is paid at the end of each fraction 1/m of a year on £1 invested at the start of the year

Consider on a timeline

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Nominal Interest Rates

- For every £1 you invest, you earn $£\frac{i^{(m)}}{m}$ after a period of 1/*m.*

- For every £P you invest, you earn $£P\frac{i^{(m)}}{m}$ after a period of 1/*m.*

- Accumulated value of £1 invested at *t* = *1* is $\left(1+\frac{i^{(m)}}{m}\right)^{m}$.

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Nominal Interest Rates

If *i* is the effective annual interest rate, then:

$$1 + i = \left(1 + \frac{i^{(m)}}{m}\right)^m$$

More generally:

$$\left(1 + \frac{i^{(p)}}{p}\right)^p = \left(1 + \frac{i^{(m)}}{m}\right)^m$$

Where p and m are positive rational numbers

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Some examples

For the effective rates shown below, calculate the nominal rate per annum:

Effective rate (p.a.)	Nominal rate p.a., compounding half-yearly	Nominal rate p.a., compounding monthly
$i = 0.06$	$i^{(2)} =$	$i^{(12)} =$
$i = 0.12$		
$i = 0.126835$		
$i = 0.092025$		

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More examples

For the nominal rates shown below, calculate the following:

Nominal rate per annum	Nominal rate p.a., compounding monthly	Effective rate of interest per annum	Effective rate of interest per month
$i^{(4)} = 0.12$	$i^{(12)} =$	$i=$	
$i^{(6)} = 0.06$	$i^{(12)} =$	$i=$	
$i^{(0.5)} = 0.10$	$i^{(12)} =$	$i=$	
$i^{(1)} = 0.08$	$i^{(12)} =$	$i=$	

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Nominal Rates of Discount

- Recall that $d = 1 - v = iv$
- Interpretation: d = the amount of interest payable at the beginning of the year that is equivalent to i paid at the end of the year.
- The nominal discount rate $d^{(m)}$ represents interest of $d^{(m)}/m$ paid at the start of each 1/m of a year

$$1 - d = \left(1 - \frac{d^{(m)}}{m}\right)^m.$$

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Nominal discount examples

Complete the following table:

Effective rate of discount p.a.	Nominal rate of discount, convertible monthly, p.a.	Nominal rate of discount, convertible half-yearly, p.a.
$d = 0.04$	$d^{(12)} =$	$d^{(2)} =$
$d =$	$d^{(12)} = 0.06$	$d^{(2)} =$
$d =$	$d^{(12)} =$	$d^{(2)} = 0.10$

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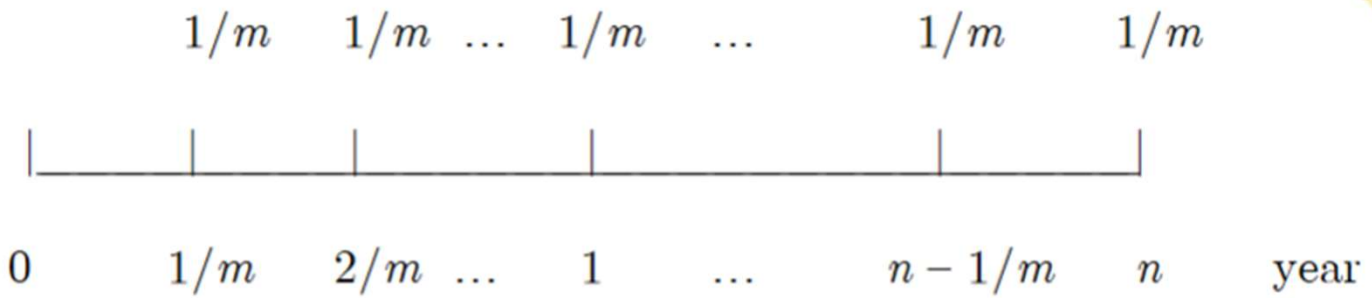
Annuities payable *m*thly – new notation

- $a_{\overline{n}|}^{(m)}$: The value at time zero of a level annuity payable $m$thly in arrears at the rate of 1 per unit time over the time interval $[0, n]$.
- $a_{\overline{n}|}^{(m)}$: The value at time zero of a level annuity which pays $1/m$ in arrears m times per year for a period of n years.
- $s_{\overline{n}|}^{(m)}$: The value at time n of a level annuity payable $m$thly in arrears at the rate of 1 per unit time over the time interval $[0, n]$.

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Consider an annuity paid *m times each year*

$\text{PV} = a_{\overline{n}|}^{(m)}$ $\qquad$ $\text{AV (at } t = n) = s_{\overline{n}|}^{(m)}$



PV and FV of that annuity paid *m times each year*

$$
\begin{aligned}
a_{\overline{n}|}^{(m)} &= \tfrac{1}{m} v^{\frac{1}{m}} + \tfrac{1}{m} v^{\frac{2}{m}} + \ldots + \tfrac{1}{m} v^{\frac{mn-1}{m}} + \tfrac{1}{m} v^{\frac{mn}{m}} \\
&= \frac{v^{\frac{1}{m}}(1-(v^{\frac{1}{m}})^{nm})}{m(1-v^{\frac{1}{m}})} \\
&= \frac{1-v^n}{m\left[(1+i)^{\frac{1}{m}}-1\right]} \\
&= \frac{1-v^n}{i^{(m)}} \\
&= \frac{i}{i^{(m)}} a_{\overline{n}|}
\end{aligned}
$$

$$
s_{\overline{n}|}^{(m)} = (1+i)^n a_{\overline{n}|}^{(m)} = \frac{(1+i)^n - 1}{i^{(m)}} = \frac{i}{i^{(m)}} s_{\overline{n}|}
$$

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PV and FV of that annuity-due paid *m* *times each year*

$$
\begin{aligned}
\ddot{a}_{\overline{n}|}^{(m)} &= \tfrac{1}{m} + \tfrac{1}{m} v^{\frac{1}{m}} + \tfrac{1}{m} v^{\frac{2}{m}} + \ldots + \tfrac{1}{m} v^{n-\frac{1}{m}} \\
&= \frac{1-(v^{\frac{1}{m}})^{nm}}{m(1-v^{\frac{1}{m}})} \\
&= \frac{1-v^n}{m[1-(1+i)^{-\frac{1}{m}}]} \\
&= \frac{1-v^n}{d^{(m)}} \\
&= \frac{i}{d^{(m)}} a_{\overline{n}|} \\
&= \frac{d}{d^{(m)}} \ddot{a}_{\overline{n}|}
\end{aligned}
$$

$$
\begin{aligned}
\ddot{s}_{\overline{n}|}^{(m)} &= (1+i)^n \ddot{a}_{\overline{n}|}^{(m)} \\
&= \frac{(1+i)^n - 1}{d^{(m)}} \\
&= \frac{i}{d^{(m)}} s_{\overline{n}|} \\
&= \frac{d}{d^{(m)}} \ddot{s}_{\overline{n}|}
\end{aligned}
$$

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Useful results (that you should be able to prove from first principles)

$$
\begin{aligned}
a_{\overline{n}|}^{(m)} &= \sum_{t=1}^{nm} \frac{1}{m} v^{\frac{t}{m}} = \frac{1-v^n}{i^{(m)}} \\
\ddot{a}_{\overline{n}|}^{(m)} &= \sum_{t=1}^{nm} \frac{1}{m} v^{\frac{t-1}{m}} = \frac{1-v^n}{d^{(m)}} \\
a_{\overline{\infty}|}^{(m)} &= \frac{1}{i^{(m)}} \\
\ddot{a}_{\overline{\infty}|}^{(m)} &= \frac{1}{d^{(m)}} \\
s_{\overline{n}|}^{(m)} &= \sum_{t=1}^{nm} \frac{1}{m} (1+i)^{n-\frac{t}{m}} = \frac{(1+i)^n - 1}{i^{(m)}} \\
\ddot{s}_{\overline{n}|}^{(m)} &= \sum_{t=1}^{nm} \frac{1}{m} (1+i)^{n-\frac{t-1}{m}} = \frac{(1+i)^n - 1}{d^{(m)}}
\end{aligned}
$$

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Annuity Example 1

Calculate the present value as at 1 January 2014 of the payments of £100 made on the first day of each quarter during the calendar years 2014 and 2015, if the effective rate of interest is 8% per annum.

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Annuity Example 2

A company expects to receive £10,000 from a customer. The amount will be split into ten equal instalments. The first instalment is due in one years' time and the remaining payments will be made at three-year intervals thereafter. Calculate, to the nearest £, the present value of these future receipts at an effective rate of 7% per annum.

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Attendance matters!

